

Md Rahatul Ashakin

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Research Focus

My research focuses on quantum machine learning and advanced AI for solving challenges in healthcare, biology, and secure data systems. I am passionate about developing innovative, interdisciplinary methods that enhance prediction, diagnosis, and data-driven discovery while ensuring strong security. I seek to collaborate across domains and apply new technologies to impactful problems in science and society.

Research and Professional Experience

Data Science Intern, Tekurai Inc., San Antonio, Texas (remote, Queens - NY) *May 2024 to Present*

- Built transformer and CNN pipelines for clinical text and audio, analyzed clinician and patient interactions for decision support.
- Processed more than five hundred thousand healthcare records with Python and produced interactive Power BI dashboards for operations and care insights.
- Prototyped privacy aware blockchain workflows for secure data exchange in drug discovery across multiple institutions.
- Quantum machine learning with Qiskit, including hybrid variational models and quantum guided hyperparameter search that improved internal accuracy and AUC over classical baselines.

Program Manager, ShopUp, Dhaka, Bangladesh *Feb 2022 to Sep 2022*

- Led analytics for B2C and B2B operations with forecasting and KPI reporting; delivered Tableau and Power BI dashboards used to reduce returns and improve throughput.
- Partnered with operations, product, and engineering to define a KPI taxonomy and the supporting SQL/ETL pipelines; automated weekly executive dashboards and cohort analyses to speed up decision cycles.

Project Manager, Icon Information Systems Ltd., Dhaka, Bangladesh *Jun 2020 to Feb 2022*

- Directed cross-team reporting and SQL data workflows for enterprise clients; coordinated agile ceremonies and risk tracking.
- Produced project plans and stakeholder updates, organized sprints and stand-ups, and prepared UAT and release checklists to ensure on-time, auditable deliveries.

Assistant Project Manager, Millennium Information Solution Ltd., Dhaka, Bangladesh *Nov 2018 to May 2020*

- Supported delivery of HRIS and financial systems with user acceptance testing, reliability checks, and transaction safety reviews for regulated stakeholders.

Education

M.S., Information Technology, Washington University of Science and Technology (WUST), Alexandria, Virginia *Oct 2022 to Mar 2024*

B.S., Computer Science and Engineering, North South University, Dhaka, Bangladesh *Sep 2013 to Aug 2018*

Publications & Scholarly Contributions

Book Chapter (forthcoming)

- **Ashakin, M. R.** (2026). Chapter 5: *Current Tools and Techniques of Artificial Intelligence in Healthcare*. In D. Roosan (Ed.), *Quantum Computing in Medicine: Transforming Drug Discovery, Protein Folding, and Genomics Research through Artificial Intelligence Integration*. Elsevier. ISBN: 978-0-443-34193-9. *CReditT*: Conceptualization; Methodology; Writing: original draft; Visualization; Writing: review & editing.

Under Review (Journal)

- **Ashakin, M. R.**, Stuetzle, C., Khan, R., Nirzhor, S., Roosan, D. *Quantum Annealing of Financial Sentiment Analysis by Optimizing Large Language Model Hyperparameters*. Under review at *Financial Innovation (SpringerOpen)*. *CReditT*: Lead: Conceptualization; Methodology; Software; Formal analysis; Visualization; Writing: original draft. Supporting: Data curation; Writing: review & editing. *Resources*: Code: [GitHub repository](#).

Refereed Conference Papers

- Roosan, D., **Ashakin, M. R.**, Khan, R., Hasiba Mashed Khan, Khou, T., Carnasciali, M. I., Haider, M. R. (2025). *Variational Quantum Circuits for Molecular Classification Using Graph Neural Network*. In: *IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC 2025)*. *CReditT*: Methodology; Software; Formal analysis; Visualization; Writing: review & editing.
- Roosan, D., Khan, R., **Ashakin, M. R.**, Khou, T., Nirzhor, S., Haider, M. R. (2025). *Quantum Variational Transformer Model for Enhanced Cancer Classification*. In: *IEAI 2025, IOS Press, Advances in Transdisciplinary Engineering* *CReditT*: Methodology; Software; Validation; Formal analysis; Writing: review & editing.
- Roosan, D., Khan, R., **Ashakin, M. R.**, Khou, T. (2025). *Adaptive Multimodal Artificial Intelligence with Liquid Neural Network for Edge Computing-Based Augmented Reality*. In: *IEAI 2025, IOS Press, Advances in Transdisciplinary Engineering*. *CReditT*: Conceptualization; Methodology; Software; Visualization; Writing: review & editing.
- Roosan, D., Khou, T., Pham, B., Li, Y., **Ashakin, M. R.**, Hasiba Mashed Khan, Khan, R., Haider, M. R. (to appear, Nov 2025). In: *Quantum AI Enhanced Blockchain Security for Drug Discovery*. *Proceedings of ICAI 2025, Springer LNEE (to appear)*. *CReditT*: Conceptualization; Methodology; Writing: review & editing.

Journal Articles

- **Ashakin, M. R.**, Bhuyian, M. S., Hosain, M. R., Deya, R. S., Hasan, S. E. (2024). *Transforming to Smart Healthcare: AI Innovations for Improving Affordability, Efficiency, and Accessibility*. *Pathfinder of Research*, 2024. *CReditT*: Lead: conceptualization; methodology; writing: original draft; visualization; writing: review & editing.
- Hossain Ifty, **Ashakin, M. R.**, S. M., Hossain, B. Islam Tusher, M. B., Haque, R. (2023). *Adoption of IoT in Agriculture — Systematic Review*. *Applied Agriculture Sciences*, 2023. *CReditT*: Writing: review & editing; literature curation; visualization.
- **Ashakin, M. R.**, Hossain, B., Hossain Ifty, S. M. (2024). *Enhancing Digital Signal Processing: Obstacles and Advancements in FPGA and ASIC Hardware Implementations*. *Pathfinder of Research*, 2024. *CReditT*: Lead: conceptualization; writing: original draft; methodology; visualization; writing: review & editing.
- Hossain Ifty, S. M., **Ashakin, M. R.**, Hossain, B., Afrin, S., Sattar, A., Chowdhury, R., Tusher, M. I., Bhowmik, P. K., Mia, M. T., Islam, T., Tufael, M., Sunny, A. R. (2023). *IOT-Based Smart Agriculture in Bangladesh: An Overview*. *Applied Agriculture Sciences*, 2023. *CReditT*: Writing: review & editing; visualization; literature curation.
- Salam, M. T., Bari, K. F., Rahman, M. M., Gafur, D. M. M., Faruk, M. O., Akter, K., Irin, F., **Ashakin, M. R.**, Shaikat, T. A., Das, A. C., Tufael, M., Mithun, M. M., Sunny, A. R. (2024). *Emergence of Antibiotic-Resistant Infections in ICU Patients*. *Journal of Angiotherapy*, 2024. *CReditT*: Writing: review & editing; visualization.

- **Ashakin, M. R.**, Hasan, M. R. K., Urbi, S. R. C. (2024). *Impact of Internet of Things (IoT) on Healthcare in Transforming Patient Care and Overcoming Operational Challenges*. *Journal of Angiotherapy*, 2024. *CRedit*: Lead: conceptualization; writing: original draft; visualization; writing: review & editing.
- **Ashakin, M. R.**, Hasan, M. R. K., Urbi, S. R. C. (2024). *AI-Powered Methods for Smarter Decisions in Automated Machine Learning in Business Analytics*. *Pathfinder of Research*, 2024. *CRedit*: Lead: conceptualization; writing: original draft; methodology; visualization; writing: review & editing.
- Hossain Ifty, S. M., Irin, F., Shovon, M. S. S., Amjad, M. H. H., Bhowmik, P. K., Ahmed, R., **Ashakin, M. R.**, Hossain, B., Mushfiq, M., Sattar, A., Chowdhury, R., Sunny, A. R. (2024). *Advancements, Applications, and Future Directions of Artificial Intelligence in Healthcare*. *Journal of Angiotherapy*, 2024. *CRedit*: Writing: review & editing; visualization.

Skills

Methods & Algorithms	Transformers · CNNs · GNNs · multimodal learning · optimization for robustness · model calibration · privacy-aware modeling
Data & Experimentation	Dataset curation · experimental design · cross-validation · statistical testing · ablation studies · error analysis · reproducible pipelines
Software & Systems	Python · SQL · Git · Linux · Jupyter · Pandas · NumPy · scikit-learn · PyTorch · Hugging Face
Visualization & Reporting	Matplotlib · Tableau · Power BI · results dashboards · academic writing with LaTeX/Overleaf
Computing Platforms	GPUs · cloud · edge/IoT deployment

Selected Research Projects (open-source)

- **IBD-Quantum-ESM**. Quantum-informed *IBD* modeling pipeline (optimization with classical + quantum methods). *Resources*: Code: [GitHub](#) · [README](#) · [Notebooks](#).
- **Quantum Optimization Sandbox**. Lightweight scaffold to *repurpose* ML experiments with QAOA/VQE and classical baselines. *Resources*: Code: [GitHub](#) · [README](#) · [Notebooks](#).

Online Courses & Certifications (selected)

- [Microsoft Power BI Data Analyst Specialization](#) · **Microsoft**
- [Google Advanced Data Analytics](#) · **Google**
- [Google Business Intelligence](#) · **Google**
- [Google Data Analytics](#) · **Google**
- [SAS Statistical Business Analyst](#) · **SAS**
- [Project Management & Other Tools for Career Development](#) · **UC Irvine**

Academic Service

Research Programmer, Merrimack College, North Andover, Massachusetts (Remote, Queens - NY) *Sep 2024 to Dec 2024*

- Supported *DSE 4900: Data Science Capstone (Fall 2024)* and *DSE 3010: Machine Learning for Data Science (Fall 2024)*; coordinated labs and weekly research clinics, and supervised capstone theses end-to-end (scoping, dataset curation, modeling plans, final reports/presentations).
- Instituted reproducible research practice (clean repos, environment pinning, experiment tracking) with rigorous evaluation (ablations, error analysis, checks for leakage/imbalance/overfitting), enabling teams to achieve *up to ~50% relative accuracy gains* over course baselines where applicable.